

RAG APPROACH ON SPIN

EXTRACTION OF SPM-TARGET PROTEIN INTERACTIONS FROM PUBMED ARTICLES

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ABSTRACT

Specialized pro-resolving mediators (SPMs) are bioactive lipid mediators that actively regulate the resolution of inflammation and are increasingly studied for their therapeutic significance in chronic inflammatory and immune-mediated disorders [4, 5, 6]. Despite growing interest, the literature surrounding SPMs remains fragmented across multiple nomenclatures, synonyms, and article collections, making systematic retrieval and downstream analysis difficult.

This project presents a domain-specific Retrieval-Augmented Generation (RAG) pipeline designed to extract SPM-protein interactions from biomedical literature. The system integrates synonym expansion via PubChem, large-scale literature retrieval from PubMed and PubMed Central. Furthermore, large language model (LLM)-based relation extraction is employed to identify interaction evidence, which is subsequently represented as a graph in Neo4j.

One important aspect of this work is that it focuses only on a specific biomedical domain. Instead of relying purely on what the model already knows, the system uses actual retrieved research papers as evidence. This makes the results more reliable and easier to verify. This enables reliable literature exploration, traceable question answering, and structured knowledge extraction within the SPM domain. The results demonstrate that the proposed pipeline can effectively transform unstructured biomedical publications into a searchable and interpretable knowledge resource, providing a practical foundation for evidence-driven scientific investigation and interaction-centered knowledge discovery.

Keywords: Retrieval-Augmented Generation, Specialized Pro-Resolving Mediators (SPMs), PubMed Mining, cosine similarity, semantic search, FAISS, Mistral AI, Neo4j

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1 INTRODUCTION

The rapid growth of biomedical literature has created both opportunities and challenges for researchers. With large numbers of scientific articles published across databases such as PubMed, identifying precise, relevant, and contextually meaningful information has become increasingly difficult and time-consuming. Traditional keyword-based search methods are often insufficient because they rely on exact term matching and fail to capture semantic relationships expressed through varied phrasing, abbreviations, and chemical synonyms.

This challenge is particularly evident in the study of Specialized Pro-Resolving Mediators (SPMs), a class of bioactive lipid mediators that actively regulate the resolution of inflammation and contribute to tissue homeostasis [4, 5, 6]. SPMs, including resolvins, protectins, maresins, and lipoxins, have gained increasing importance in biomedical research due to their role in immune regulation, tissue repair, and chronic disease treatment. However, literature related to SPMs is fragmented across multiple naming conventions and research contexts, making systematic retrieval difficult.

Conventional literature search is therefore not enough when the objective is not only to find papers, but also to identify relevant evidence passages, support traceable question answering, and extract structured biological relationships such as SPM–protein interactions. In this context, Retrieval-Augmented Generation (RAG) provides a useful technical framework. RAG combines parametric language modeling with non-parametric external knowledge retrieval, thereby improving grounding, provenance, and adaptability in knowledge-intensive tasks [1]. In biomedical settings, where factual correctness and evidence traceability are especially important, such an approach is particularly suitable.

This project presents a domain-specific RAG pipeline for SPM-related biomedical literature mining. The system integrates synonym expansion through PubChem, literature retrieval from PubMed and PubMed Central, structured storage in MongoDB, semantic indexing with FAISS, citation-grounded retrieval through a Streamlit chatbot, LLM-based extraction of SPM–protein interactions, and graph-based representation in Neo4j. Beyond implementation, the project also examines practical issues encountered during development, including synonym ambiguity, incomplete full-text access, retrieval filtering trade-offs, and consistency challenges in interaction extraction.

The main objectives of this work are:

- to automate the expansion of SPM search terms using PubChem synonyms,
- to collect and organize SPM-related PubMed literature in a structured local database,
- to build a dense retrieval index for semantic literature access,
- to provide a citation-grounded chatbot for question answering over the dataset,
- to extract SPM–protein interactions with evidence sentences using an LLM, and
- to visualize the resulting interaction network in Neo4j.

The scope of the project is limited to a domain-specific biomedical retrieval and extraction pipeline for SPM-related literature. It does not aim to build a general biomedical question-answering system over all medical literature. Instead, it focuses on a narrower but meaningful domain in which the benefits of semantic retrieval, evidence grounding, and structured knowledge extraction can be examined in depth.

2 LITERATURE SURVEY

The conceptual foundation of the present work lies at the intersection of inflammation-resolution biology, retrieval-augmented language modeling, semantic vector search, and biomedical relation extraction. RAG was formalized by Lewis et al. as a hybrid paradigm in which a generative model is augmented with retrieved external knowledge, thereby improving factuality, provenance, and adaptability for knowledge-intensive language tasks [1]. One of the key advantages of the RAG framework is that the external knowledge base can be updated independently of the model weights. This is especially valuable in scientific domains where the literature changes continuously and where provenance is essential.

Dense semantic retrieval in modern RAG systems typically depends on sentence or passage embeddings. Sentence-BERT (SBERT) introduced siamese and triplet-network-based sentence embeddings that made semantic similarity search computationally practical while preserving strong performance on semantic textual similarity tasks [2]. In projects such as the present one, embedding-based retrieval is preferable to pure lexical matching because biomedical concepts often vary in wording and abbreviation. Efficient storage and search over dense vectors is commonly handled through approximate nearest-neighbor libraries such as FAISS. Johnson, Douze, and Jégou demonstrated that FAISS enables large-scale similarity search through optimized index structures and efficient vector retrieval mechanisms [3]. This makes FAISS a practical choice for local, research-oriented biomedical RAG pipelines where cost, speed, and reproducibility are important.

The biomedical domain provides particularly strong motivation for retrieval grounding. Literature-based question answering over biomedical corpora must cope with terminology variation, overlapping abbreviations, inaccessible full texts, and evidence traceability requirements. In this project, these challenges are amplified by the chemistry-driven synonym complexity of SPMs.

On the biological side, the SPM field has established that molecules such as resolvins, protectins, maresins, and lipoxins actively orchestrate the resolution phase of inflammation rather than serving merely as passive markers [6, 4, 5]. Reviews by Serhan and colleagues, Julliard et al., and Fredman and Serhan show that SPMs are implicated in immune regulation, tissue repair, pain control, and multiple inflammatory disease mechanisms [6, 4, 5]. These reviews also illustrate why structured literature mining is necessary: the field is biologically rich, clinically relevant, and distributed across many research contexts.

Compared with standard literature search tools, the present project contributes a unified applied pipeline that integrates synonym-aware retrieval, document indexing, citation-grounded question answering, evidence-based relation extraction, and graph representation. The system is not yet a fully evaluated or formally tested chatbot, but it illustrates how a RAG-based approach can

be developed and applied to support research in this domain.

3 SYSTEM ARCHITECTURE

The overall architecture follows five major phases: Data Retrieval, Data Preparation, Semantic indexing, Retrieval-Augmented Inference, and interaction graph generation.

3.1 HIGH-LEVEL PIPELINE

The system is designed as a modular and layered pipeline in which each stage can be developed, tested, and refined independently while preserving a consistent end-to-end data flow. At a high level, the pipeline begins with a curated list of canonical Specialized Pro-Resolving Mediators (SPMs), expands these through PubChem-based synonym retrieval, collects relevant literature from PubMed and PubMed Central, preprocesses and stores the resulting corpus in MongoDB, builds a semantic retrieval index using dense embeddings and FAISS, and finally supports both citation-grounded question answering and structured interaction extraction.

The implemented workflow can be summarized as follows:

1. Read the curated list of canonical SPMs from Excel.
2. Expand each SPM through PubChem synonym retrieval.
3. Query PubMed using canonical names and synonym-augmented search strings.
4. Retrieve metadata, abstracts, PMCID links, and open-access full texts.
5. Store SPM entities, papers, and mention mappings in MongoDB.
6. Build a dense retrieval index over filtered text chunks using Sentence Transformers and FAISS.
7. Use the index to support a Streamlit-based chatbot with source display and similarity scores.
8. Retrieve relevant chunks per SPM and prompt Mistral to extract SPM–protein interactions with evidence.
9. Persist extracted interactions and load the resulting network into Neo4j.

Figure 1 presents the complete high-level architecture of the system. The left side of the workflow focuses on knowledge acquisition, where canonical SPM names are enriched with PubChem synonyms to improve biomedical search coverage. The right side represents literature retrieval from PubMed, including abstracts and available full-text content. These documents are then passed through preprocessing steps such as cleaning, normalization, and chunking before being stored in MongoDB and prepared for semantic indexing. In the lower part of the pipeline, embeddings are generated and stored in a vector index, enabling efficient retrieval of semantically relevant evidence. Finally, the retrieved chunks are supplied to the Mistral model to produce evidence-grounded responses and to support structured extraction of SPM–protein interactions.

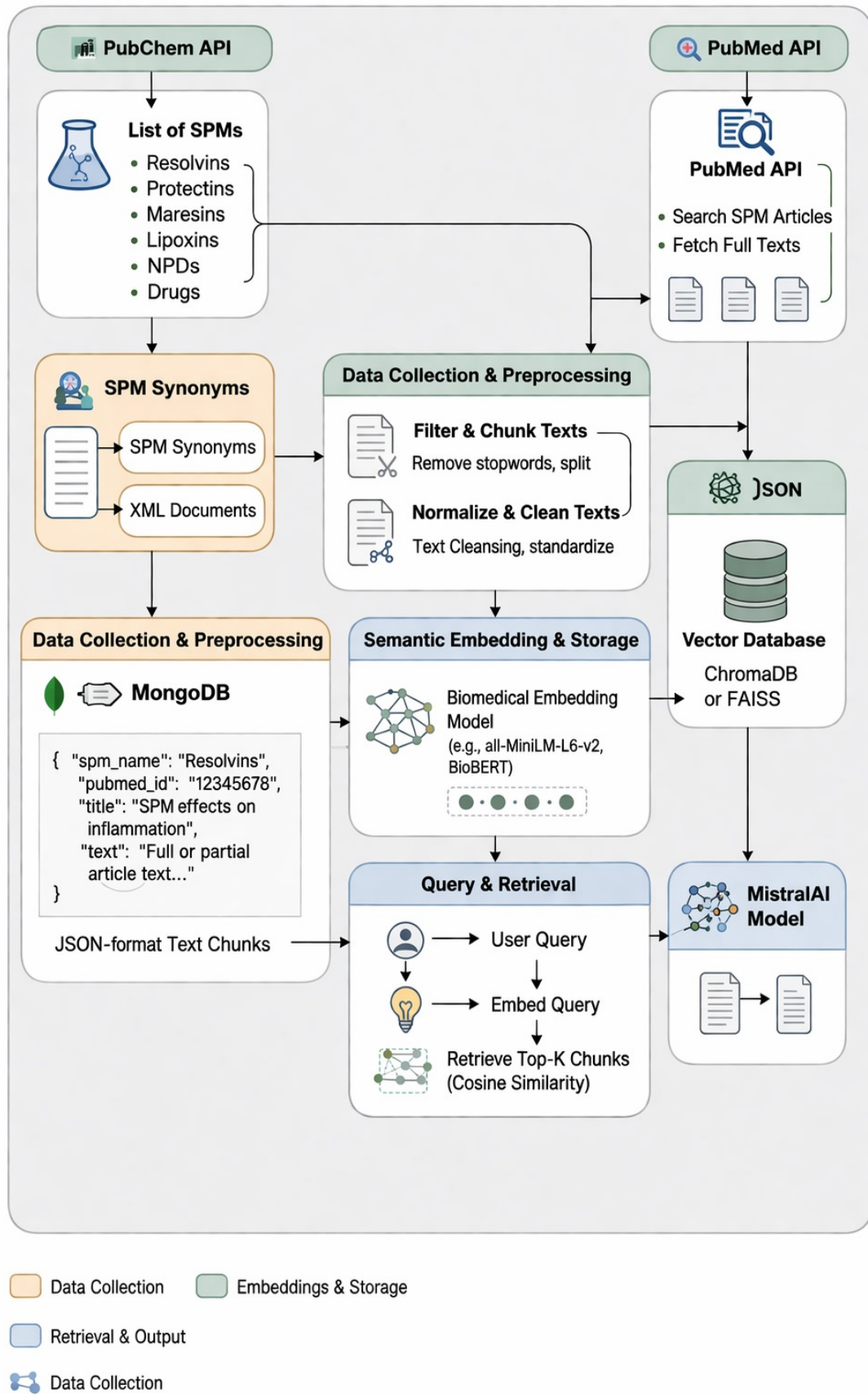


Figure 1: High-level architecture of the SPM RAG pipeline.

3.2 SYSTEM DESIGN

The architecture combines database-backed reproducibility with lightweight semantic retrieval. MongoDB serves as the central structured store for papers, SPM metadata, and mention traces, while FAISS serves as the semantic access layer for dense retrieval. The retrieval layer is coupled to a Streamlit interface that exposes source metadata and similarity scores, and the interaction extraction stage extends the system from document retrieval toward structured knowledge discovery. The LLM is not used as a standalone knowledge source; instead, it is grounded in retrieved context. This design follows the core RAG principle of augmenting parametric generation with external evidence [1]. Figure 2 illustrates the overall workflow of the proposed SPM RAG pipeline.

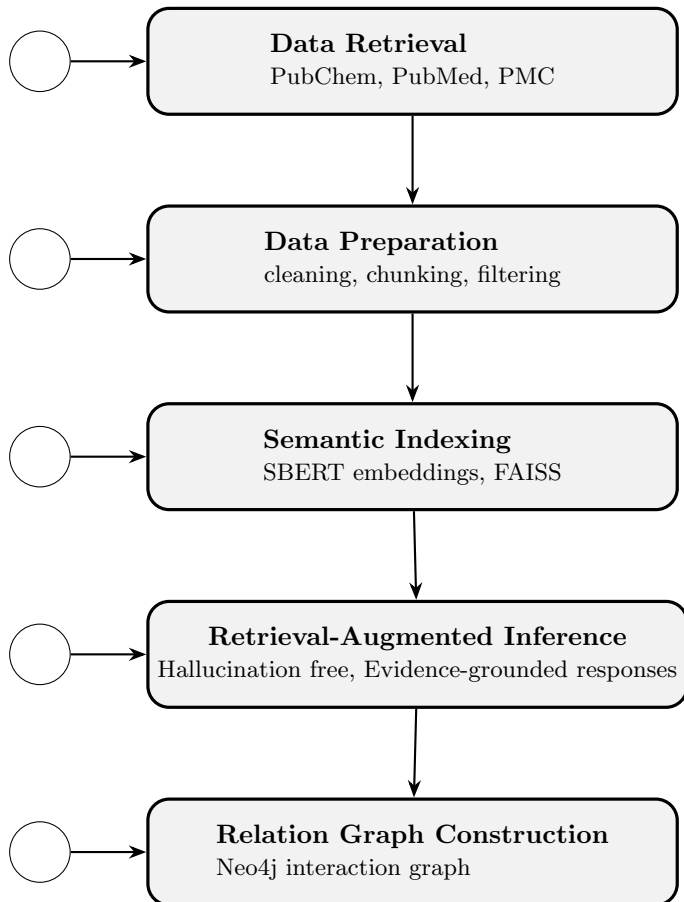


Figure 2: SPM RAG workflow with vector icons.

4 METHODOLOGY

4.1 PHASE 1: SPM SYNONYM EXTRACTION

The first step of the project focused on collecting synonyms for each SPM using the PubChem PUG REST API. The input list of SPMs was read from an Excel file, and basic cleaning was performed to remove incomplete or incorrect entries. For each valid SPM, the system queried PubChem to retrieve associated synonyms.

This step was important because SPMs are often referred to using different chemical names,

abbreviations, or alternative labels in the literature. By expanding the search terms, the system was able to improve coverage and retrieve a broader set of relevant papers.

In total, 57 SPMs across 11 classes were processed, with an average of around 12 synonyms per SPM. This highlights the importance of synonym expansion in biomedical text mining tasks.

4.2 PHASE 2: PUBMED FULL-LENGTH PAPER RETRIEVAL (PMC AND DOI-BASED)

The second stage focused on retrieving literature from PubMed using the NCBI Entrez E-utilities API [7]. For each SPM, the script combined a base query provided for the project with additional terms derived from the PubChem synonym expansion. Using this combined query, PMIDs were collected through `esearch`. Metadata and abstracts were then downloaded with `efetch`, publication dates were checked using `esummary`, and available PMCID links were obtained through `elink`. To keep the process manageable for SPMs with large result sets, a limit of 2,000 recent papers per SPM was applied.

When a PMCID was available, the system attempted to download the full-text article from PubMed Central. If the article was not available in PMC, the pipeline tried to retrieve an open-access version using the paper’s DOI. Whenever possible, PDF files were processed using PyMuPDF, while HTML versions were handled using BeautifulSoup.

This stage populated three MongoDB collections:

- **SPMs**: canonical SPM entities and metadata,
- **Papers**: paper-level metadata, abstract, and full text,
- **Mentions**: mappings between SPMs, matched synonyms, and PMIDs.

4.3 PHASE 3: DENSE INDEXING

The indexing stage transformed the stored paper corpus into a dense semantic retrieval resource. Papers were loaded from MongoDB, and the text selection strategy preferred full text when `full_text_status = OK`; otherwise, the abstract was used; if neither was available, the title served as the fallback.

Texts were chunked at paragraph level with a minimum chunk length threshold. In addition, the index builder filtered out low-value chunks and retained only those likely to contain biologically meaningful interactions, based on manually defined protein and interaction hint patterns. This filtering step helped reduce noise, especially from generic or bibliographic passages.

Embeddings were generated with the Sentence Transformers model `all-MiniLM-L6-v2`, which is built on the SBERT family of sentence embedding approaches [2]. The generated vectors were stored in a FAISS index using inner-product search over normalized embeddings, which approximates cosine similarity [3]. Alongside the vector index, metadata and original text chunks were serialized for later retrieval.

4.4 PHASE 4: RETRIEVAL-AUGMENTED INFERENCE

A Streamlit application was implemented to provide an interactive multi-chat interface for querying the SPM literature corpus. User questions were embedded using the same embedding model as the corpus index. The FAISS index returned top-k similar chunks, and the retrieved context was inserted into a prompt instructing the Mistral model to answer using only the provided evidence.

The interface stored user and assistant turns, exposed source metadata in a collapsible section, and displayed similarity scores for retrieved sources. These scores are important for transparency because they provide a visible retrieval signal to the user. However, they should be interpreted carefully: they indicate semantic proximity in embedding space rather than end-to-end answer correctness.

4.5 PHASE 5: LLM-BASED INTERACTION EXTRACTION

The final stage aimed to convert unstructured literature evidence into structured interaction records. For each canonical SPM, the system retrieved relevant chunks from the FAISS index and constructed a prompt asking the Mistral model to extract the SPM, the target protein, the relation type, the evidence sentence, the PMID, and source identifiers.

The output was parsed as JSON, normalized into a tabular structure, and stored in MongoDB. The implementation attempted to reduce duplicates via a compound unique key over SPM, protein, relation, and PMID.

4.6 PHASE 6: NEO4J KNOWLEDGE GRAPH CONSTRUCTION

To support graph-oriented exploration, a Neo4j loader script was implemented. It created unique constraints for papers, SPMs, proteins, and evidence nodes, then projected paper nodes and interaction records into a graph structure containing **SPM**, **Protein**, **Paper**, and **Evidence** labels. The graph is useful for visually identifying highly connected SPMs, shared protein targets, and evidence-linked interaction patterns.

5 RESULTS

5.1 DATA PREPARATION RESULTS

The actual exported execution artifacts show that the project achieved substantial corpus coverage, as summarized in Table 1.

The results in Table 1 show that the corpus grew from a relatively small canonical SPM list into a large paper collection. At the same time, the later stages became progressively more selective: only 51 SPMs appeared in mention records, 40 were represented in the final index, and 37 yielded extracted interactions. This reflects the effect of retrieval filtering, chunk usefulness constraints, and LLM extraction behavior.

Metric	Value
Canonical SPMs stored	57
SPM classes represented	11
Average synonyms per SPM	11.79
Unique papers stored	9,664
Papers with abstracts	9,139
Papers with DOI	9,379
Papers with PMCID	5,900
Papers with retrieved full text	5,180
Mention records stored	388,346
Unique synonym matches	687
Indexed chunks in FAISS	54,040
Unique PMIDs represented in the index	7,964
SPMs represented in mentions	51
SPMs represented in the final index	40
Extracted interaction records	140
Unique proteins in interactions	61
Unique PMIDs in interactions	54
SPMs represented in interactions	37

Table 1: Summary statistics of the executed pipeline.

5.2 MONGODB DATA STORAGE

To ensure structured storage and efficient retrieval, the collected and processed data were stored in MongoDB using multiple collections.

```

_id: ObjectId('697a66becc02292cd0c72c08')
SPM: "15-epi-LXA4"
Relation: "binds"
PMID: "12806051"
Protein: "ALXR"
Evidence: Array (1)
  0: "Among eicosanoids, ALXR is stereoselective for LXA4 (5S,6R,15S-trihydr..."
created_at: "2026-01-28T19:42:54.639293"
run_id: "20260128_204243"

_id: ObjectId('697a66becc02292cd0c72c09')
SPM: "15-epi-LXA4"
Relation: "binds"
PMID: "37686088"
Protein: "FPR2/ALX"
Evidence: Array (2)
  0: "LXs act mainly through binding to their receptors. FPR2/ALX, a G prote..."
  1: "FPR2/ALX, a G protein-coupled receptor, is the predominant LX receptor..."
created_at: "2026-01-28T19:42:54.649915"
run_id: "20260128_204243"

_id: ObjectId('697a66c2cc02292cd0c72c0a')
Protein: "A549 lung cancer cells"
PMID: "9535221"
Relation: "inhibits"
SPM: "15-epi-LXB4"
Evidence: Array (1)
  0: "..."
created_at: "2026-01-28T19:42:58.746052"

```

Figure 3: MongoDB interaction collection showing extracted SPM–protein relationships with associated evidence and source references.

Figure 3 shows a sample of the interaction collection, which contains extracted SPM–protein relationships. Each record includes the SPM, target protein, relation type (e.g., binds, inhibits), supporting evidence sentences, and the corresponding PMID. This demonstrates that the system captures not only relationships but also their supporting context.

ADD DATA EXPORT DATA UPDATE DELETE 25 1 - 25 of 9664
<pre> _id: ObjectId('696d4380cc02292cd0c1194e') pmid: "40736329" abstract: "This study aimed to investigate whether Maresin conjugates in tissue r..." created_at: 2026-01-18T20:33:04.021+00:00 doi: "10.1097/WNR.0000000000002198" doi_url: "https://doi.org/10.1097/WNR.0000000000002198" full_text: "" full_text_source: "NONE" full_text_status: "NO_OA_FOUND" full_text_url: "" journal: "Neuroreport" pmc_url: "" pmcid: "" pubmed_url: "https://pubmed.ncbi.nlm.nih.gov/40736329/" title: "MCTR1 alleviates remifentanyl-induced hyperalgesia by regulating mitoc..." year: "2025" </pre>
<pre> _id: ObjectId('696d4380cc02292cd0c1195c') pmid: "40615878" abstract: "Malaria-associated acute respiratory distress syndrome (MA-ARDS) is a ..." created_at: 2026-01-18T20:33:04.045+00:00 doi: "10.1186/s12936-025-05462-y" doi_url: "https://doi.org/10.1186/s12936-025-05462-y" full_text: "Background Malaria-associated acute respiratory distress syndrome (MA-..." full_text_source: "PMC" full_text_status: "OK" full_text_url: "https://pmc.ncbi.nlm.nih.gov/articles/PMC12228167/" journal: "Malaria journal" pmc_url: "https://pmc.ncbi.nlm.nih.gov/articles/PMC12228167/" </pre>

Figure 4: MongoDB papers collection storing PubMed metadata, including abstracts, DOIs, and full-text availability status.

Figure 4 presents the papers collection, which stores metadata retrieved from PubMed, including abstracts, DOIs, publication sources, and full-text availability. The presence of different full-text status values highlights real-world challenges such as limited open-access availability.

ADD DATA EXPORT DATA UPDATE DELETE 25 1 - 25 of 57
<pre> _id: ObjectId('696d4372cc02292cd0c11915') name: "MCTR1" created_at: 2026-01-18T20:32:50.846+00:00 downloaded: "3D" pubchem_id: "122368871.0" pubchem_link: "https://pubchem.ncbi.nlm.nih.gov/compound/122368871" pubchem_synonyms_raw: "MCTR1; 1784701-61-6; (4Z,7Z,9E,11E,13R,14S,16Z,19Z)-13-[(2R)-2-[[[(4S)-" reference: "https://portlandpress.com/essaysbiochem/article/64/3/443/226308/Specia..." spm_class: "Cys-SPMs" </pre>
<pre> _id: ObjectId('696d4372cc02292cd0c11916') name: "MCTR2" created_at: 2026-01-18T20:32:50.849+00:00 downloaded: "3D" pubchem_id: "122368872.0" pubchem_link: "https://pubchem.ncbi.nlm.nih.gov/compound/122368872" pubchem_synonyms_raw: "MCTR2; 1784701-62-7; (4Z,7Z,9E,11E,13R,14S,16Z,19Z)-13-[(2R)-2-amino-3-" reference: "https://portlandpress.com/essaysbiochem/article/64/3/443/226308/Specia..." spm_class: "Cys-SPMs" </pre>
<pre> _id: ObjectId('696d4372cc02292cd0c11917') name: "MCTR3" created_at: 2026-01-18T20:32:50.851+00:00 downloaded: "3D" pubchem_id: "122368873.0" pubchem_link: "https://pubchem.ncbi.nlm.nih.gov/compound/122368873" pubchem_synonyms_raw: "MCTR3; 1784701-63-8; orb1741501; SCHEMBL29620679; CHEBI:138209; AKOS04..." </pre>

Figure 5: MongoDB SPM collection containing PubChem-enriched mediator data, including synonyms, Pubchem identifiers, links and spm classification.

Figure 5 illustrates the SPM collection, which contains canonical mediators enriched with PubChem information, including chemical identifiers, synonyms, and classification. This step enables synonym-aware retrieval and improves coverage of the literature search.

Together, these collections form the backbone of the system, supporting downstream tasks such as semantic retrieval, question answering, and interaction extraction.

5.3 FULL-TEXT RETRIEVAL OUTCOMES

Full-text acquisition proved to be one of the most practically challenging stages of the project. Although 9,664 papers were collected, only 5,180 had retrieved full text available in the ex-

port. The status distribution in Table 2 highlights the difficulty of legal and technical access to biomedical literature.

Full-text status	Count
OK	5,180
NO_OA_FOUND	742
FETCH_ERROR	1,129
UNPAYWALL_ERROR	563
NO_DOI	58
PENDING	1,992

Table 2: Distribution of full-text retrieval status values in the exported paper collection.

Full-text source	Count
PMC	5,900
DOI_OA_PDF	121
DOI_OA_HTML	288
NONE	3,355

Table 3: Full-text source categories recorded in the paper collection.

Table 2 and Table 3 justify the practical observation in the project notes that full-paper acquisition was time-consuming and often incomplete because of access restrictions and open-access availability.

5.4 SEMANTIC RETRIEVAL AND CHATBOT BEHAVIOR

The implemented chatbot demonstrates that the indexed corpus can be queried interactively. The interface shown in Figure 6 includes a working Streamlit application titled *SPM RAG Pipeline*, support for multiple chats, answers to questions such as “What is SPM?” and “What is the role of SPM in osteoarthritis?”, citation-style source references in generated answers, an expandable “Sources used” section, and visible similarity scores for retrieved sources.

This is a meaningful result because it confirms that the project is not limited to offline indexing; rather, it supports interactive literature exploration. The UI demonstrates provenance-oriented behavior, which is particularly valuable in biomedical contexts where unsupported claims are unacceptable.

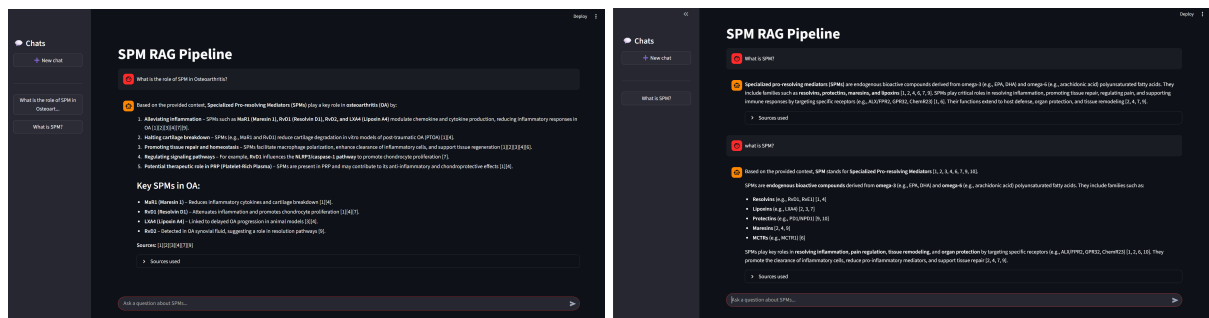


Figure 6: The implemented SPM RAG chatbot showing citation-grounded answers and source disclosure.

5.5 INTERACTION EXTRACTION RESULTS

The interaction extraction stage produced 140 interaction records across 37 SPMs, 61 proteins, and 54 PMIDs. The exported MongoDB interaction collection appears to aggregate two execution batches, as indicated by two distinct run identifiers. Since no separate run log was available, these were treated as successive extraction runs rather than fully independent experiments.

Relation label	Count
binds	55
modulates	33
activates	18
inhibits	13
upregulates	8
downregulates	7
stimulates	3
unknown	2
regulates	1

Table 4: Distribution of extracted relation labels in the interaction collection.

The relation distribution shown in Table 4 indicates that *binds*, *modulates*, and *activates* are the most frequent labels. This suggests that the extracted interaction space is centered on receptor-level or signaling-linked evidence. This is consistent with the biological literature on SPMs, where receptor engagement and downstream immunomodulation are frequent discussion points [4, 5].

5.6 GRAPH DATABASE RESULTS

Neo4j was used as a graph-based representation of the extracted knowledge. At the time of the provided screenshot, the graph overview displayed 9,949 nodes and 888 relationships. The graph included SPM, Protein, Paper, and Evidence node categories. Additional screenshots shown in Figure 7 presented both a global interaction view and focused subgraphs, for example around RvD1 and LXA4.

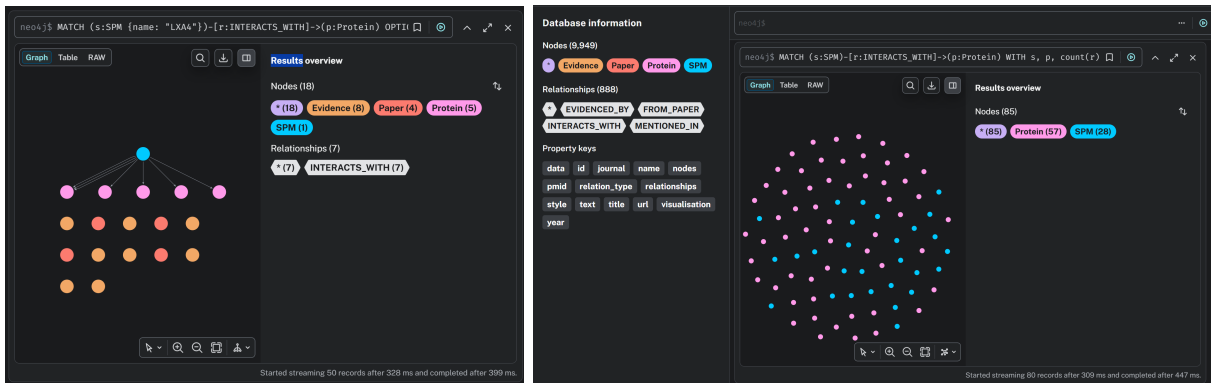


Figure 7: Neo4j visualization of extracted SPM–protein interaction structures.

The graph visualization is impactful because it transforms text-derived evidence into an interpretable network representation. A focused subgraph, such as the RvD1 neighborhood, is

especially suitable for the report because it clearly shows one SPM node linked to several protein targets without the clutter of the full graph.

6 CHALLENGES

The development of the system involved several practical challenges across different stages of the pipeline. A key challenge during data collection was synonym ambiguity. While synonym expansion improved retrieval coverage, not all synonyms were equally useful. Certain terms were overly broad or contextually ambiguous, leading to irrelevant results and requiring iterative filtering and manual refinement. In addition, inconsistencies in PubMed data, including variation in XML structures and incomplete article availability, required the implementation of fallback strategies and flexible parsing methods.

Preprocessing introduced a critical trade-off between noise reduction and context preservation. Strict filtering risked removing relevant information when SPMs were referenced indirectly, whereas relaxed filtering increased noise and reduced retrieval precision. Similarly, chunk size selection required balancing context retention with retrieval specificity. Larger chunks preserved more context but reduced precision, while smaller chunks improved precision but risked losing important information.

Further challenges arose in embedding generation and retrieval ranking. Model selection involved balancing computational efficiency with semantic accuracy, and similarity-based ranking did not always prioritize the most informative results. Maintaining consistent metadata linkage between retrieved text segments and source documents was essential for interpretability. Additionally, ensuring that the frontend interaction layer remained grounded in retrieved evidence, while still providing readable outputs, required careful prompt design and validation. Evaluation also posed challenges, as defining meaningful test queries and interpreting metric outputs required both quantitative and qualitative analysis.

7 LIMITATIONS

Despite the successful implementation of the system, several limitations remain. Full-text availability is incomplete, as a substantial portion of retrieved articles provide only abstracts or partially structured content. This limits the contextual depth of the data and affects downstream stages such as preprocessing, chunking, embedding generation, and evidence extraction. In addition, the indexed dataset represents a filtered subset of the collected articles. While this improves efficiency and reduces noise, it may exclude relevant information that is expressed in less explicit or indirect forms.

Interaction extraction is limited in both coverage and consistency. Only a subset of SPMs produced structured interaction outputs, and variation in relation labels such as *stimulates* and *regulates* introduces inconsistencies in the resulting graph representation. Retrieval quality is also dependent on the selected embedding model, where general-purpose models may not fully capture domain-specific biomedical semantics, while specialized models introduce computational complexity.

Finally, evaluation remains constrained. Although RAG-based metrics provide useful indicators of relevance and faithfulness, they do not fully capture scientific correctness. In biomedical contexts, automated evaluation must be complemented with expert validation to ensure reliability and interpretability of the results.

8 FUTURE WORK

Several directions can further enhance the system in terms of robustness, accuracy, and usability. A primary area for improvement is the introduction of more rigorous evaluation. Future work should incorporate expert-reviewed datasets, benchmark queries, and extended RAG-based metrics such as faithfulness, answer relevance, context precision, and context recall to provide a more comprehensive assessment of system performance.

The interaction extraction stage can be improved through ontology-constrained prompting and stricter normalization of entities and relation labels. This would improve consistency and enable more reliable graph construction. Retrieval quality can be further enhanced through hybrid approaches that combine semantic embeddings with lexical matching and metadata-based filtering. The use of domain-specific biomedical embedding models and reranking techniques may also improve precision, particularly for complex or low-frequency queries.

In addition, the Neo4j layer can be extended beyond visualization toward analytical applications, including centrality analysis, clustering, and domain-specific subnetwork exploration. The frontend can be improved by introducing richer citation display, explicit confidence indicators, and user-controlled retrieval parameters such as top-k selection and content filtering. Continuous data updating and integration with newly published literature would further transform the system into a dynamic research support tool.

9 CONCLUSION

This project demonstrates that a domain-specific biomedical literature-mining workflow can be effectively transformed into a Retrieval-Augmented Generation system. By integrating synonym expansion, large-scale literature retrieval, structured storage, preprocessing, semantic indexing, retrieval-grounded interaction, and graph-based representation, the system supports both exploratory access and structured analysis of biomedical knowledge.

A key strength of the system is its emphasis on evidence grounding and transparency. Retrieved information is linked to source documents and presented with supporting text, ensuring traceability and improving reliability in biomedical contexts. The system also highlights the effectiveness of semantic retrieval over traditional keyword-based approaches, particularly in domains characterized by complex terminology and multiple naming conventions.

At the same time, the project reveals practical limitations and challenges related to data availability, preprocessing trade-offs, embedding model selection, retrieval quality, and evaluation complexity. Overall, the system serves as a strong proof of concept and establishes a solid foundation for future improvements in domain-specific RAG systems and biomedical information retrieval.

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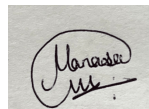
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I hereby declare by my own signature that I completed this paper independently and without the use of any sources or aids other than those stated. All passages taken directly or indirectly from published sources have been identified as such. The work has not previously been submitted in the same or a similar form for assessment or recognition. I am aware that submitting a false declaration may result in the examination being considered failed.

Rostock

March 31, 2026

(Submission Date)

A handwritten signature in black ink, appearing to read 'Manasvi', is shown within a rectangular frame.

(Full Signature)